

Lab 2 – TrashTag Product Specification

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1 Introduction

Addressing the prevalent litter crisis affecting environments and communities worldwide remains a persistent threat that evokes immediate intervention. Over the past 36 years, volunteers in the state of Virginia have removed approximately 7.1 million pounds of litter (Chesapeake Bay Foundation, 2025), requiring hundreds of people to thoroughly scan affected areas for trash (Fawaz, 2023). The environmental repercussions are severe, as illegal dumping has led to substantial marine wildlife mortality (Campbell, 2016). Property values have also been negatively impacted by illegal dumping, as it has been shown to progressively weaken the house market by 7%-10% (City of Hampton, 2025). As a result, communities continue to suffer the environmental and economic consequences of waste dumping, with tourism revenue decreasing by an astounding 38% (Keep Texas Beautiful, 2024).

TrashTag is a browser application developed to address the specific communication concerns causing perpetual inefficiency in cleanup responses due to the absence of real-time coordination between citizens that encounter litter and cleanup organizations that acquire the resources to remove it. The interactive live map provides users with the capability to take geotagged pictures and immediately make that data available to cleanup organizations capable of resolving the task. The systems innovative features facilitate communication between litter-reporting citizens and cleanup organizations, creating a surge in efficient cleanups and positive impact on environmental restoration efforts.

1.1 Purpose

The software requirements specification (SRS) serves to introduce the functional and non-functional requirements for the TrashTag prototype, which is intended for developers and testers who will ensure the browser application is functioning optimally, facilitating the implementation process. The prototype will diminish the existent communication barrier between citizens reporting a litter incident and clean up organizations, as well as enhance the environmental health of continuously affected communities.

1.2 Scope

TrashTag is a browser application that drives positive environmental impact by providing an enhanced, structured communication pathway for litter-reporting citizens to efficiently interact with cleanup organizations and volunteers. The platform will allow citizens to upload a geotagged image as organizations provide live updates on cleanup progress for the problematic areas. The prototype will include a reporting feature in which geotagged images can be successfully uploaded to a live interactive map displaying real-time litter locations, while the platform continuously updates cleanup progress to maintain accurate status tracking. The group scheduler, analytics dashboard, and municipal waste systems integration are part of the platform's future implementation plan and therefore, will not be included in the prototype.

1.3 Definitions, Acronyms, and Abbreviations

Gamification: The incorporation of game-like mechanics such as rankings to motivate user participation and increase cleanup operations.

Geotagging: The addition of GPS coordinates to images that enable precise location identification where trash was reported.

Illegal Dumping: The unauthorized disposal of trash and other hazardous materials onto private and or public property.

Live Real-Time Updates: Live data synchronization displaying trash reports to both users and cleanup organizations instantaneously.

Metadata: Descriptive information within data files such as data, location, and timestamp data in photos to be submitted to TrashTag.

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1.5 Overview

This software requirement specification has been segmented into two categories, each with its respective subsections detailing certain technical aspects of the TrashTag platform. Section 2, however, contains a higher concentration of technical depth as it encompasses the products perspective, functions, user characteristics, constraints, and assumptions and dependencies of the prototype.

2 Overall Description

TrashTag will be implemented as a browser-based application driven by the interaction of two target groups, users that encounter litter masses and report that data and organizations that acquire the resources and staff to resolve the prevalent littering issue. The core features such as uploading reports with geotagged images and dynamic status updates will be included in this prototype, while all other features will be incorporated with future demos due to current constraints.

2.1 Product Perspective

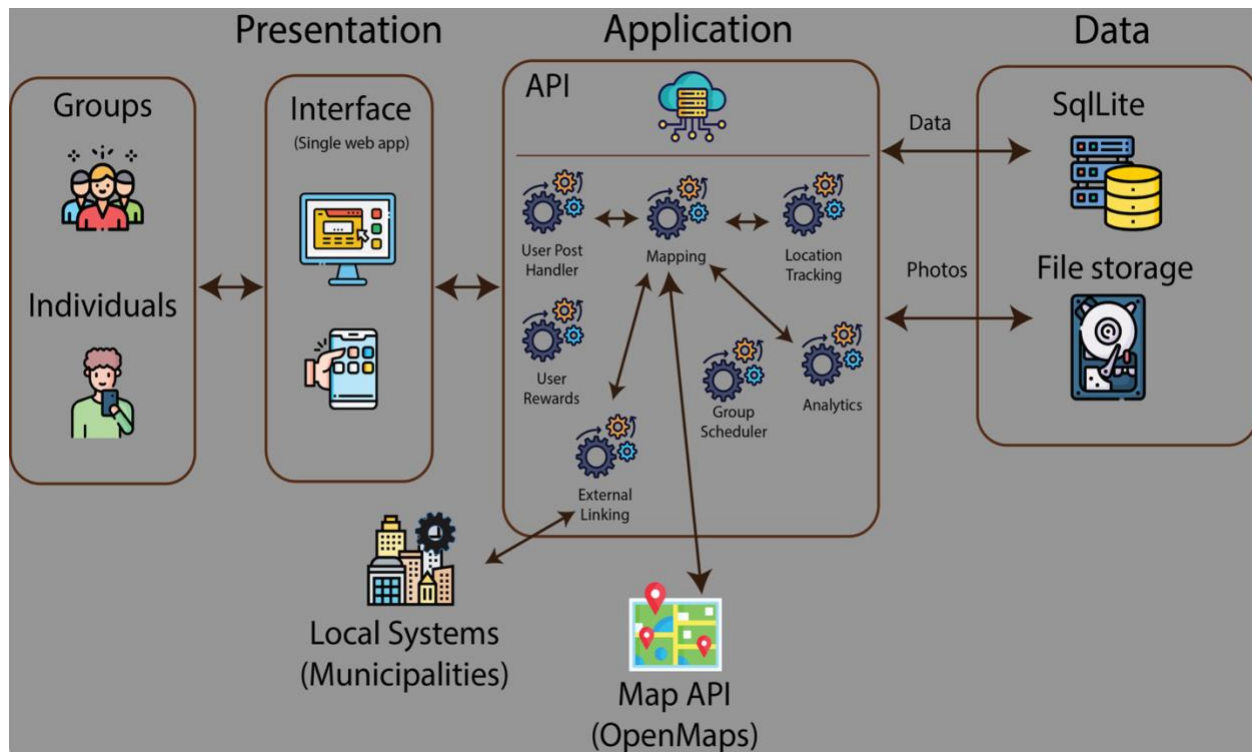
TrashTag will act as a standalone browser application designed to engage environmentally driven communities to resolve their communication issues associated with problematic litter areas. Hence, this application is independent and not dependent on larger systems. Given that

this platform hosts a three-tier architecture system as demonstrated in figure 1, as opposed to the traditional web development approach, React will manage the user interface components such as report submissions, real-time interactive maps, and accurate status tracking to both desktop and mobile browsers. Whereas Django Python will focus solely on the backend logic functioning as a REST API handling litter report information and coordination between litter reporters and cleanup crews.

The data storage layer will operate using SQLite to store all information pertaining to the user reports, accurate status updates, and communication between the users, while photo uploads will be stored separately in a file storage system due to SQLite's inability to efficiently store large images. Nginx acts as a middleman, receiving all incoming traffic for the React server side, forwarding and receiving any HTTPS API requests and responses to and from Gunicorn. Since Nginx is unable to decipher Python, it sends all API requests to Gunicorn, as it efficiently interprets Python logic. Initially, Nginx was incorporated for React, however, the geolocation feature remained non-functional as Django continued to process HTTP protocols without SSL further confusing the browser. Hence, SSL incorporation was necessary to enable HTTPS across React and Django to prevent further browser confusion. The system will integrate OpenMaps API to render geolocation updates on the interactive map to allow users to view the most recent advancements in status tracking.

Figure 1

TrashTag Prototype Major Functional Component Diagram



2.2 Product Functions

The browser application supports the following functions:

Table 1

TrashTag Feature Description and Prototype Implementation

Feature	Description	Prototype Implementation
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Report Litter Location	Users can submit location of encountered litter-infested area to the live interactive map.	Fully Implemented
Upload Photos	Users can capture images of the litter-infested area at the time of report submission.	Fully Implemented
Mark Trash Size(LBS)	Users can give an estimate of the encountered litter mass in pounds.	Fully Implemented
Request Cleanup Verification Status	Users can verify if the reported litter area has been cleared.	Fully Implemented
View Litter Map	Live litter reports are displayed on the live interactive map.	Fully Implemented
Litter Density Map	Map can display areas with high litter report traffic.	Eliminated - Time constraints
Filter by Status	Users can filter litter reports depending on the cleanup status.	Eliminated- deemed nonessential
Filter by Location	Users can filter litter reports depending on the specific location.	Fully Implemented

Coordinate Cleanup Events	Organizations can allocate resources and staff to participate in cleanup events.	Partial Implementation- Basic Demo
Claim Cleanup Area	Organizations can claim a reported littered area to clean with their staff/resources.	Partial Implementation- Basic Demo
Delegate Cleanup Tasks	Organizations can assign and manage certain cleanup tasks to their staff and volunteers.	Partial Implementation- Basic Demo
View Cleanup Events	Users can view the displayed cleanup events within near proximity on the interface.	Partial Implementation- Basic Demo
RSVP to Events	Users can reserve a spot to participate in cleanup events.	Partial Implementation- Basic Demo
Check-In to Events	Users can check in to events to track progress and involvement which will then be reflected in gamification rewards.	Partial Implementation- Basic Demo
Proximity Cleanup Alerts	Users can be notified when there are cleanup events in the area.	Partial Implementation- Basic Demo

Update Cleanup Status	Volunteers can update the status of a litter-reported area to in progress or cleaned.	Fully Implemented
Upload Cleanup Proof	Volunteers can capture an image directly from the platform to provide evidence that area has been cleaned.	Fully Implemented
Monitor Cleanup Status	Users can monitor the live status of a litter report on the platform.	Fully Implemented
Verify Accurate Cleanup Status	To ensure cleanup status is accurate, the system requires all images to be captured directly from the platform.	Fully Implemented
Verify Cleanup Completion	To ensure cleanup completion, system must receive images directly through the platform at update submission.	Fully Implemented
Flag Suspicious Content	Users can flag litter-reports as suspicious or threatening.	Eliminated- Deemed Nonessential
View Impact Statistics	Users can view statistics on how cleanup efforts have affected communities and the environment.	Partial implementation – Basic Stats

Gamifications/Rewards	Users can earn points that go towards their ranking in the system depending on their cooperation and cleanup efforts.	Fully Implemented
Top Contributors	A dashboard displaying the most active participants in the application, whether it be users creating litter reports or volunteers cleaning up said areas.	Fully Implemented
Local Laws and Resources	Users can access platform guidelines.	Partial implementation – Basic Stats
Embed Map Widget	Interactive map can be imbedded into certain platforms.	Partial implementation – Basic Stats
Highlight Predicted Litter Trends	Analytics dashboard predicating areas most likely to become litter-infested due to past trends.	Eliminated- Deemed Nonessential
Download Reports	Users can download litter report data for educational and informational purposes.	Eliminated- Deemed Nonessential

View Density Heavy Litter Analytics	Users can view a dashboard displaying patterns of litter-heavy areas.	Eliminated- Deemed Nonessential
Track Event Impact	Analytics dashboard can track the impact cleanup events have on the environment and communities.	Eliminated- Deemed Nonessential
View Cleanup Statistics	Users can view cleanup statistics.	Eliminated- Deemed Nonessential
Review Flagged Reports	Administrators can review flagged reports created by user roles.	Eliminated- Deemed Nonessential
Respond to Flagged Reports	Administrators can respond to flagged reports created by user roles.	Eliminated- Deemed Nonessential
Edit/Correct Flagged Report Data	Administrators can edit or correct flagged data that has been deemed incorrect.	Eliminated- Deemed Nonessential
Remove Suspicious Reports	Administrators can remove reports that have been flagged as suspicious.	Eliminated- Deemed Nonessential
Mark Reports as Spam	Administrators can mark litter reports as spam.	Eliminated- Deemed Nonessential

Issue Violation Warnings	Administrators can issue violation warnings for any platform violation.	Eliminated- Deemed Nonessential
Create New Account	Users can create new account to create and view other litter reports on the platforms.	Fully Implemented
Login	Users can login to their accounts on the platform.	Fully Implemented
View and Edit Profile	Users can view and edit their profile information.	Fully Implemented

2.3 User Characteristics

TrashTag is designed to support three user role types comprising citizens, who implement the reporting aspect into the system, organizations, who resolve the litter reported area issue, and local government, who optimize environmental cleanup efforts through the data analytics provided by the platform. Citizens are not required to attain or sustain technical literacy for this system, however, basic browser navigation would suffice to accurately report and upload geotagged images to the interactive map. Subsequently, as Organizations encompass more tasks and responsibilities in the interface, a more intermediate technical level of expertise is required to coordinate resources and allocate staff while updating the status of the littered area to maintain accuracy on the platform. Local government users do not require an extensive level of

proficiency to manage the interface given that the sole objective is to analyze data to formulate better policy and budget decisions while all user information remains protected and concealed.

2.4 Constraints

The prototype functions under several constraints ranging from technical to legal factors. An early complication that arose as the production environment was being established, was the specific security permission feature that required user bypassing to save and update their geolocation data onto the interactive map. Yet, if the permission were to be denied then the geolocation features would be unable to process the litter report information.

Aside from the client-side rendering hurdle, the geolocation feature introduced a new SSL constraint, which prevented the server from continuing its normal operations. Since the browser required HTTPS over HTTP protocol for higher security encryption and verification, React and Django would no longer satisfy these demands as they functioned under HTTP. Hence Nginx was introduced to support React server-side rendering, handling all requests from the user interface, while Gunicorn operated as a Python server to execute Django dependencies, finalizing HTTPS request and response protocols exchanged between them. Although all actions are daemonized, if either Nginx or Gunicorn crash then the whole platform becomes dysfunctional, as React would be unable to send API calls to Django with a nonfunctioning Nginx and Gunicorn standing between them. If OpenMaps API malfunctions, the interactive map feature crashes along with the service as it is highly dependent on it. The incorporation of such servers will prevent any data leakage as SSL ensures encryption and security of user information stored in the database. Additionally, the legal constraints demand an End-User License Agreement

(EULA) to profoundly state TrashTag is not responsible for any casualties sustained through the platform.

2.5 Assumptions and Dependencies

It is assumed that all users of the TrashTag browser application will have reliable internet connection along with a stable web browser environment. It is also assumed that users have a functioning device capable of capturing images to allow photos to be taken directly for the platform onto the live map.

The system depends on certain Python libraries being installed to fully implement Django into the production environment. The system relies on OpenMaps API functionality to ensure the geolocation remains in optimal conditions.